

X has pmf/pdf $f(x|\theta): \theta \in \Theta$

Test: $H_0: \theta \in \Theta_0$ vs. $H_1: \theta \in \Theta_1$ $\Theta_0 \cap \Theta_1 = \emptyset$

power fn. $\beta_\phi(\theta) = \text{power fn of a test } \phi$
 $= E_\theta[\phi(X)]$

$\beta_\phi(\theta) = \text{power if } \theta \in \Theta_1$, $\text{Size} = \sup_{\theta \in \Theta_0} \beta_\phi(\theta)$

1. Consistency: $\beta_{\phi_n}(\theta) \rightarrow 1$ as $n \rightarrow \infty \forall \theta \in \Theta_1$
2. Unbiasedness: A test ϕ for testing $H_0: \theta \in \Theta_0$ vs. $H_1: \theta \in \Theta_1$ is unbiased if

$$\underbrace{\sup_{\theta \in \Theta_0} \beta_\phi(\theta)}_{\text{size}} \leq \underbrace{\beta_\phi(\theta)}_{\text{power}} \forall \theta \in \Theta_1$$

$\Leftrightarrow$ the chance of rejecting H_0 is higher when H_1 true than when H_0 is true.

Ex. $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma_0^2)$
 $\sigma_0^2 \rightarrow \text{known}$

$H_0: \mu = \mu_0$ vs. $H_1: \mu > \mu_0$

Show $\phi(\bar{x}) = 1 \left(\frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma_0} > z_\alpha \right)$ $\xrightarrow{T(\bar{x})}$

$$\beta_\phi(\mu) = P\left(\frac{\sqrt{n}(\bar{x} - \mu + \mu - \mu_0)}{\sigma_0} > z_\alpha \right)$$

$$= 1 - \Phi\left(z_\alpha - \frac{\sqrt{n}(\mu - \mu_0)}{\sigma_0}\right)$$

$$\geq 1 - \Phi(z_\alpha)$$

$$= \alpha$$

A biased test $\phi_1(z) = 1\left(\frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma_0} < z_\alpha\right)$

Show $\beta_{\phi_1}(\mu) < \alpha \quad \forall \mu > \mu_0$

The best test

To consider the class C of all level- α (size- α) tests and choose the one that has $\min^m P(\text{Type-II error})$, i.e. has $\max^m$ power

Defⁿ: A test ϕ^* for testing $H_0: \theta \in \Theta_0$ vs. $H_1: \theta \in \Theta_1$ ($\Theta_1 \neq \Theta_0$) is called a Most Powerful level- α (or size- α) test if ← prefixed

(MP)

$$(i) \quad \sup_{\theta \in \Theta_0} \beta_{\phi^*}(\theta) \leq \alpha \quad (= \alpha \text{ for size-}\alpha)$$

$$(ii) \quad \beta_{\phi^*}(\theta_1) \geq \beta_{\phi}(\theta_1) \quad \forall \phi \text{ satisfying (i)}$$

Defn: A test ϕ^* for testing $H_0: \theta \in \Theta_0$ vs. $H_1: \theta \in \Theta_1$, ($\Theta_1 \neq \Theta_0$) is called a Uniformly Most Powerful (UMP) level- α test if (ϕ^* is uniformly MP $\forall \theta_1 \in \Theta_1$)

$$(i) \sup_{\theta \in \Theta_0} \beta_{\phi^*}(\theta) \leq \alpha \quad (= \alpha \text{ for size-}\alpha)$$

$$(ii) \beta_{\phi^*}(\theta_1) \geq \beta_{\phi}(\theta_1) \quad \forall \theta \in \Theta_1 \\ \forall \phi \text{ satisfying (i)}$$

UMP tests does not always exist, but it exists in certain situations, such as

(1) Simple vs Simple $H_0: \theta = \theta_0$ vs. $H_1: \theta = \theta_1$ (NP Lemma)

(2) $H_0: \theta = \theta_0$ vs. $H_1: \theta > \theta_0$ (Application of NP Lemma)
(or $< \theta_0$)

(3) $H_0: \theta \leq \theta_0$ vs. $H_1: \theta > \theta_0$ (MLR)

1. To find UMP tests, we should look for tests that are functions of sufficient stat.

If S is suff for $\theta \in \Theta$ Consider $\phi_1(x) = E(\phi(x) | S)$

$$\text{power fn. } \beta_{\phi_1}(\theta) = E_{\theta}[\phi_1(x)] = E_{\theta}[E(\phi(x) | S)]$$

$$= E_{\theta} \phi(x)$$

$$= \beta_{\phi}(\theta)$$

2. UMP level- α tests should be unbiased.

Consider $\phi_0(x) = \alpha \quad \forall x$ (also a level α test)

Suppose ϕ^* UMP level- α . Then

$$\text{power of } \phi^* = \beta_{\phi^*}(\theta) \geq E_{\theta}(\phi_0(x)) = \alpha \geq \sup_{\theta \in \Theta_0} \beta_{\phi^*}(\theta)$$

Any statistic $\phi: \mathcal{X} \rightarrow [0, 1]$ is a test fn.

If $X = \underline{x}$ $\phi(\underline{x})$ = prob with which we reject H_0

Neyman-Pearson Fundamental Lemma (only sufficiency part with non-randomized test)

Suppose $X = (X_1, \dots, X_n)$ has pmf/pdf $f(x|\theta)$; $\theta \in \Theta \subseteq \mathbb{R}^k$. Consider testing $H_0: \theta = \theta_0$ vs. $H_1: \theta = \theta_1$ (for known $\theta_0 \neq \theta_1$). For prefixed $0 < \alpha < 1$ if ϕ^* is a test fn. such that

$$\phi^*(x) = \begin{cases} 1 & \text{if } f(x|\theta_1) > K f(x|\theta_0) \\ 0 & \text{if } f(x|\theta_1) < K f(x|\theta_0) \end{cases} \quad (1)$$

where $K (\geq 0)$ is a constant determined from $E_{\theta_0}[\phi^*(X)] = \alpha$, then ϕ^* is a MP size- α test for H_0 vs. H_1 $\leftarrow (2)$

Pf: Suppose ϕ^* is a test that satisfies both (1) and (2) and ϕ is any test that only satisfies (2). Suppose R^* and R be the rejection regions of ϕ^* and ϕ resp.

$$R^* = \{x: \phi^*(x) = 1\} = \{x: f(x|\theta_1) > K f(x|\theta_0)\}$$

$$R = \{x: \phi(x) = 1\}$$

$$\text{power difference } \beta_{\phi^*}(\theta_1) - \beta_{\phi}(\theta_1)$$

$$= \int_{R^*} f(x|\theta_1) dx - \int_R f(x|\theta_1) dx$$

$$= \int_{R^* \setminus (R \cap R^*)} f(x|\theta_1) dx + \int_{R \cap R^*} f(x|\theta_1) dx - \int_{R \setminus R \cap R^*} f(x|\theta_1) dx - \int_{R \cap R^*} f(x|\theta_1) dx$$

$$\geq \int_{R^* \setminus (R \cap R^*)} K f(x|\theta_0) dx - \int_{R \setminus (R \cap R^*)} K f(x|\theta_0) dx \quad (\text{change of measure})$$

$$= K \left(\int_{R^* \setminus (R \cap R^*)} f(x|\theta_0) dx + \int_{R \cap R^*} f(x|\theta_0) dx - \int_{R \setminus (R \cap R^*)} f(x|\theta_0) dx - \int_{R \cap R^*} f(x|\theta_0) dx \right)$$

$$= K \left(\underbrace{\int_{R^*} f(x|\theta_0) dx}_{\alpha} - \underbrace{\int_R f(x|\theta_0) dx}_{\alpha} \right)$$

$$= 0$$

$$\Rightarrow \beta_{\phi^*}(\theta_1) \geq \beta_{\phi}(\theta_1)$$

Note K is obtained by solving $P_{\theta_0}(f(x|\theta_1) > K f(x|\theta_0)) = \alpha$

Ex 1 $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma_0^2)$ Test $H_0: \mu = \mu_0$ vs. $H_1: \mu = \mu_1$
 $\xrightarrow{\text{known}} \quad \xrightarrow{\text{known}}$

Find a MP Size- α test for H_0 vs H_1 (Case A: $\mu_1 > \mu_0$)
 (Case B: $\mu_1 < \mu_0$)

$$\frac{f(x|\mu_1)}{f(x|\mu_0)} = e^{-\frac{1}{2\sigma_0^2} \left\{ \sum_{i=1}^n \left[(x_i - \mu_1)^2 - (x_i - \mu_0)^2 \right] \right\}}$$

$$= \cancel{(n-1)s^2} + n(\bar{x} - \mu_1)^2 - \cancel{(n-1)s^2} - n(\bar{x} - \mu_0)^2$$

$$= e^{-\frac{n}{2\sigma_0^2} \left\{ (\bar{x} - \mu_1)^2 - (\bar{x} - \mu_2)^2 \right\}}$$

$$g(\bar{x}) = \mathbb{P} \left(\frac{n(\mu_1 - \mu_0)}{2\sigma_0^2} (2\bar{x} - \mu_0 - \mu_1) > K \right)$$

$= R$ (rejection region)

Case A: $(\mu_1 > \mu_0)$

g is increasing fn.

$$\frac{f(x|\mu_1)}{f(x|\mu_0)} > K \iff \bar{x} > c$$

Find c : $P_{\mu_0}(\bar{X} > c) = \alpha$

$$P_{\mu_0} \left(\underbrace{\frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma_0}}_{N(0,1)} > \underbrace{\frac{\sqrt{n}(c - \mu_0)}{\sigma_0}}_{z_\alpha} \right) = \alpha$$

$\therefore$ The MP Size- α test is

$$\begin{aligned} \phi(x) &= \mathbb{1} \left(\frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma_0} > z_\alpha \right) \\ &= \mathbb{1} \left(\bar{x} > \mu_0 + \frac{\sigma_0}{\sqrt{n}} z_\alpha \right) \end{aligned}$$

Case B: $(\mu_1 < \mu_0)$

g is decreasing fn.

$$\{x: g(\bar{x}) > K\} = \{x: \bar{x} < c\}$$

Find c : $P_{\mu_0}(\bar{X} < c) = \alpha \Rightarrow P_{\mu_0} \left(\frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma_0} < \underbrace{\frac{\sqrt{n}(c - \mu_0)}{\sigma_0}}_{-z_\alpha} \right)$

$\Rightarrow$ NP Size- α test :

$$\phi(x) = \mathbb{1}(\bar{x} < \mu_0 - \frac{\sigma_0}{\sqrt{n}} z_\alpha)$$

Ex NP Lemma when no parameter involved based on a single X . Suppose $X \sim f$

Test: $H_0: f \equiv N(0, 1/2)$ vs. $H_1: f \equiv \text{Cauchy}(0, 1)$

$H_0: X \sim N(0, 1/2)$ vs. $H_1: X \sim \text{Cauchy}(0, 1)$

Find a MP Size- α test

Consider

$$\frac{f_1(x)}{f_0(x)} = \frac{\frac{1}{\pi} \frac{1}{1+x^2}}{\frac{1}{\sqrt{\pi}} e^{-x^2}} = \frac{1}{\sqrt{\pi}} \frac{e^{x^2}}{1+x^2} > K$$

$$\Rightarrow \ln(e^{x^2}(1+x^2)^{-1}) > K_1$$

$$\Rightarrow x^2 - \ln(1+x^2) > K_1$$

$$=: g(x)$$

$$g'(x) = 2x - \frac{2x}{1+x^2} = \frac{2x^3}{1+x^2} \geq 0 \text{ if } x \geq 0$$

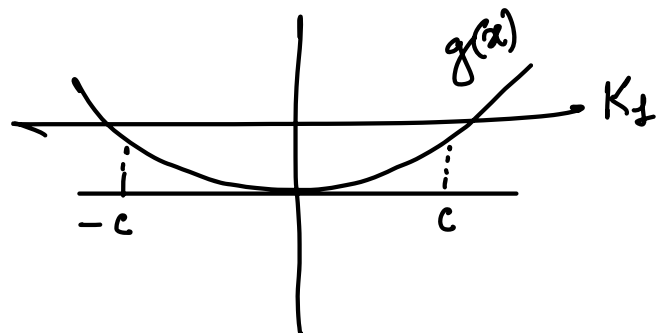
$\therefore g(x) \uparrow$ if $x > 0$
 $\downarrow$ if $x < 0$

$\therefore g(x) > K_1 \Leftrightarrow |x| > c$ for some c

Find c : $P_{f_0}(|X| > c) = \alpha$

$$\Rightarrow 2P_{f_0}(X > c) = \alpha$$

$$N(0, 1/2)$$



$$\Rightarrow P_{f_0}(\sqrt{2}X > \sqrt{2}c)$$

$\widetilde{N(0,1)} \quad \parallel \quad Z_{\alpha/2}$

$$\Rightarrow 1 - \Phi(\sqrt{2}c) = \alpha/2$$

$$\Rightarrow \sqrt{2}c = \Phi^{-1}(1 - \alpha/2) = Z_{\alpha/2}$$

$$\therefore \text{MP size-}\alpha \text{ test } \phi(X) = \mathbb{1}_{(|X| > Z_{\alpha/2}/\sqrt{2})}$$

$$\text{Power of } \phi = P_{f_1}(|X| > c)$$

$$= 2P_{f_1}(X > c)$$

$$= 2 \int_c^{\infty} \frac{1}{\pi(1+x^2)} dx$$

$$= \frac{2}{\pi} \tan^{-1} x \Big|_c^{\infty} = \frac{2}{\pi} \left(\frac{\pi}{2} - \tan^{-1} c \right)$$

$$= 1 - \frac{2}{\pi} \tan^{-1} \left(\frac{Z_{\alpha/2}}{\sqrt{2}} \right)$$

$$\approx 0.3971$$

$$\text{if } \alpha = 0.05.$$